



BARODA UNIVERSITY

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"Think Big... Think Beyond"

ITM SLS BARODA UNIVERSITY, VADODARA
B.TECH. SEMESTER-3 (BRANCH – ALL; SCHOOL – SOCSET)
DEPARTMENT OF APPLIED MATHEMATICS (S.O.S.)
COMPUTER ORIENTED NUMERICAL METHODS WITH PYTHON (S2320C1)

TUTORIAL – SOLUTION OF ALGEBRAIC & TRANSCENDENTAL EQUATIONS

Part-A: Multiple Choice Questions

- Q-1. The number of roots of a polynomial of degree 'n' has
(a) n-1 (b) n (c) n+1 (d) n+2
- Q-2. If $f(x)$ has a root between points x_1 & x_2 then $f(x_1)$ & $f(x_2)$ have signs which are
(a) negative (b) positive (c) same (d) opposite
- Q-3. The convergence of which of the following method is prone to starting value?
(a) Regula-Falsi (b) Newton-Raphson (c) Bisection (d) all
- Q-4. The Newton-Raphson method fails when
(a) $f'(x)$ is negative (b) $f'(x)$ is positive (c) $f'(x)$ is zero (d) never fails
- Q-5. A transcendental equation may contain:
(a) logarithmic function (b) exponential function (c) trigonometric function (d) all
- Q-6. One real root of the equation $x^3 - x - 1 = 0$ lies in the interval
(a) 0 and 1 (b) 1 and 2 (c) 2 and 3 (d) none of the mentioned
- Q-7. Bisection method does not work if two roots are:
(a) equal (b) unequal (c) almost equal (d) none of the mentioned
- Q-8. Lin-Bairstow's method is used to find:
(a) real roots (b) negative roots (c) positive roots (d) complex roots
- Q-9. In the iteration method for rapid convergence, the value of $\phi'(x)$ should be
(a) larger (b) smaller (c) zero (d) none
- Q-10. For a polynomial, Budan's theorem is used to estimate, within an interval:
(a) number of complex roots (b) number of real roots
(c) total number of roots (d) all

Part-B: Theory Questions

- Q-1. Define with examples:
(a) Algebraic Equation (b) Transcendental Equation
- Q-2. If the permissible error is ϵ , then what is approximate number of iterations "n" required for converging to a root in Bisection method?
- Q-3. State the type & rate/order of convergence for methods:
(a) Bisection (b) Regula-Falsi (c) Secant (d) Newton-Raphson
- Q-4. State Descartes's Rule of signs.
- Q-5. State Budan's Theorem.

- Q-6. Compare all the iterative methods- Bisection, Regula-Falsi, Secant and Newton-Raphson methods.
- Q-7. Derive the iterative formula for Newton-Raphson method.
- Q-8. State the alter names for methods:
(a) Bisection (b) Regula-Falsi (c) Secant (d) Iteration (e) Newton-Raphson
- Q-9. What is the difference between bracketing method and open method? List the methods under each type with their general iterative formula.
- Q-10. List at least two demerits of each iterative method.

Part-C: Numerical Questions

- Q-1. Using Descartes's Rule of signs, find the number of complex roots for:
(a) $2x^7 - x^5 + 4x^3 - 5 = 0$ (b) $x^7 - 3x^4 + 2x^3 - 1 = 0$
- Q-2. Using Budan's Theorem, find the number of roots of equations:
(a) $x^3 - 3x^2 - 4x + 13 = 0$ In $[-3, -2], [-2, -1], [-1, 0], [0, 1], [1, 2], [2, 3]$.
Ans: In sequence of intervals-One, zero, zero, zero, zero, either two or zero.
(b) $x^5 - x^4 + 3x^3 + 9x^2 - x + 5 = 0$ In $[-3, -2], [-2, -1], [-1, 0], [0, 1]$.
Ans: In sequence of intervals-zero, zero, one, either four or two or zero.
(c) $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1 = 0$ In $[-2, -1], [-1, 0], [0, 1], [1, 2]$.
Ans: In sequence of intervals-one, one, one, either two or zero
- Q-3. Obtain a root for the following equations using Bisection Method:
(a) $5x \log_{10} x - 6 = 0$ Correct to four significant digits.
(b) $x^2 + x - \cos x = 0$ Correct to three decimal places.
(c) $4e^{-x} \sin x - 1 = 0$ To a tolerance of 0.05%.
(d) $e^{-x} - x = 0$ To a tolerance of 0.01%.
(e) $x^3 + 4x + 9 = 0$ Correct to three decimal places.
Ans: (a) 2.741 (b) 0.550 (c) 0.370 (d) 0.567 (e) -2.7064
- Q-4. Obtain a root for the following equations using Regula-Falsi Method:
(a) $x = 3e^{-x}$ Correct to four significant digits.
(b) $x \tan x + 1 = 0$ Correct to three decimal places in $[2, 3]$
(c) $x \log_{10} x = 1.2$ To a tolerance of 0.03%.
(d) $x^{2.2} = 69$ To a tolerance of 0.01%.
(e) $x^3 - 2x - 5 = 0$ Correct to three decimal places.
Ans: (a) 1.049 (b) 2.798 (c) 2.740 (d) 6.852 (e) 2.094

Q-5. Obtain a root for the following equations using Secant Method:

- (a) Fourth root of 32 correct to four significant digits.
- (b) $x \log_{10} x = 1.9$ Correct to three decimal places.
- (c) $3x + \sin x = e^x$ To a tolerance of 0.09%.
- (d) $x^3 - 5x + 1 = 0$ To a tolerance of 0.04%.
- (e) $xe^x = 2$ Correct to three decimal places.

Ans: (a) 2.377 (b) 3.496 (c) 0.360 (d) 2.128 (e) 0.853

Q-6. Obtain a root for the following equations using Newton-Raphson Method:

- (a) $x^4 - x = 10$ correct to four significant digits.
- (b) $x \sin x + \cos x = 0$ Correct to three decimal places.
- (c) $x^{\sin^2} - 4 = 0$ To a tolerance of 0.010%.
- (d) $xe^x = \cos x$ To a tolerance of 0.02%.
- (e) $x^3 - 3x + 1 = 0$ Correct to three decimal places.

Ans: (a) 1.856 (b) 2.798 (c) 4.593 (d) 0.517 (e) 1.532

Q-7. Obtain a root for the following equations using Iteration Method:

- (a) $e^x = 3x$ correct to five significant digits.
- (b) $x^3 = 1 - x^2$ In the interval $[0,1]$ of order 10^{-4} .
- (c) $2x - \log_{10} x = 7$ To a tolerance of 0.002%.
- (d) $\cos x = 3x - 1$ To a tolerance of 0.02%.
- (e) $x^3 - 2x + 5 = 0$ Correct to four decimal places.

Ans: (a) 0.6190 (b) 0.754 (c) 3.7892 (d) 0.607 (e) -2.0945

Q-8. Obtain a quadratic factor of the form $x^2 + px + q$ for the following equations using Lin-Bairstow's Method: ***(Look for Note No. 6 at the end)**

- (a) $x^4 - 3x^3 + 20x^2 + 44x + 54 = 0$ Using $p_0 = 2$ & $q_0 = 2$
- (b) $x^4 - x^3 + 6x^2 + 5x + 10 = 0$ Using $p_0 = 1.14$ & $q_0 = 1.42$
- (c) $x^3 + 3.7x^2 + 6.25x - 4.069 = 0$ Using $p_0 = -2.5$ & $q_0 = 3$
- (d) $x^3 + x^2 - x + 2 = 0$ Using $p_0 = -0.9$ & $q_0 = 0.9$
- (e) $x^4 + x^3 + 2x^2 + x + 1 = 0$ Using $p_0 = 0.5$ & $q_0 = 0.5$ Perform one iteration.

Ans: (a) $p = 1.9413, q = 1.9538$ (b) $p = 1.1446, q = 1.4219$ (c) $p = -2.4001, q = 3.1299$
(d) $p = -1.0000, q = 1.0000$ (e) $p = 0.6667, q = 1.0000$

Q-9. Evaluate using iterative formulae from deductions of Newton's method correct up to four decimal places:

- (a) $\sqrt{21}$ (b) $\frac{1}{18}$ (c) $\frac{1}{\sqrt{15}}$ (d) $(28)^{-\frac{1}{4}}$

Ans: (a) 4.5826 (b) 0.0555 (c) 0.2582 (d) 0.4347

Note:

1. The answers may vary according to “interval” of convergence.
2. The answers may vary with respect to last digit of decimal place according to the number of fixed decimals while calculations.
3. Tolerance of 0.01%=0.0001 & 0.05%=0.0005 implies calculations up to three decimal accuracy.
4. Four significant digits implies three decimal accuracy.
5. Order $10^{-4}=0.00010$ implies three decimal accuracy.
- *6. Consider and remember the general formula for Lin-Bairstow's Method in calculations in Q-8:

$$\Delta p = \frac{b_{(n-1)}c_{(n-2)} - b_n c_{(n-3)}}{c_{(n-2)}^2 - c_{(n-3)}(c_{(n-1)} - b_{(n-1)})} \quad \text{and} \quad \Delta q = \frac{b_n c_{(n-2)} - b_{(n-1)}(c_{(n-1)} - b_{(n-1)})}{c_{(n-2)}^2 - c_{(n-3)}(c_{(n-1)} - b_{(n-1)})}$$

where n=highest order (power) of polynomial and accordingly, the table values will be used.

For (a),(b) & (e): n=4

For (c) & (d): n=3